

Environmental Data Analytics

Week 1: Data Collection & Initial Exploration

Capstone Project – EcoSphere Institute

Group Number: 14 Capstone Project – Environmental Data Analytics EcoSphere Institute

```
In [28]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [29]: # Load Datasets

temp_df = pd.read_csv("Temperature_Data.csv")
aq_df = pd.read_csv("Air_Quality_Data.csv")
geo_df = pd.read_csv("Geospatial_Data.csv")

print("Datasets loaded successfully")
```

Datasets loaded successfully

```
In [30]: #Dataset Shapes

print("Temperature Dataset Shape:", temp_df.shape)
print("Air Quality Dataset Shape:", aq_df.shape)
print("Geospatial Dataset Shape:", geo_df.shape)
```

Temperature Dataset Shape: (148, 1)
Air Quality Dataset Shape: (29531, 16)
Geospatial Dataset Shape: (6159, 4)

```
In [31]: #Dataset Column names

print("Temperature Columns:")
print(temp_df.columns.tolist())

print("\nAir Quality Columns:")
print(aq_df.columns.tolist())

print("\nGeospatial Columns:")
print(geo_df.columns.tolist())
```

Temperature Columns:
['Land-Ocean: Global Means']

Air Quality Columns:
['City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3', 'CO', 'SO2', 'O3', 'Benzene', 'Toluene', 'Xylene', 'AQI', 'AQI_Bucket']

Geospatial Columns:
['Land', 'Year', 'Series', 'Value']

```
In [32]: #DataSet Preview
print("Temperature Dataset")
display(temp_df.head())

print("\nAir Quality Dataset")
display(aq_df.head())

print("\nGeospatial Dataset")
display(geo_df.head())
```

Temperature Dataset

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	J-D	D-N	DJF
1880	-.19	-.25	-.10	-.17	-.10	-.21	-.19	-.11	-.15	-.24	-.22	-.19	-.18	***	***
1881	-.20	-.16	.02	.04	.06	-.19	.01	-.04	-.15	-.22	-.19	-.07	-.09	-.10	-.18
1882	.16	.14	.04	-.16	-.14	-.22	-.16	-.08	-.15	-.23	-.17	-.36	-.11	-.09	.08
1883	-.29	-.36	-.12	-.18	-.17	-.07	-.07	-.14	-.22	-.11	-.24	-.11	-.17	-.20	-.34

Air Quality Dataset

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzen
0	Ahmedabad	2015-01-01	NaN	NaN	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.0
1	Ahmedabad	2015-01-02	NaN	NaN	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.6
2	Ahmedabad	2015-01-03	NaN	NaN	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.8
3	Ahmedabad	2015-01-04	NaN	NaN	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.4
4	Ahmedabad	2015-01-05	NaN	NaN	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.0

Geospatial Dataset

	Land	Year	Series	Value
0	Total, all countries or areas	2005	Land area (thousand hectares)	1.299769e+07
1	Total, all countries or areas	2005	Arable land (thousand hectares)	1.356869e+06
2	Total, all countries or areas	2005	Permanent crops (thousand hectares)	1.456805e+05
3	Total, all countries or areas	2005	Forest cover (thousand hectares)	4.032743e+06
4	Total, all countries or areas	2005	Arable land (% of total land area)	1.043930e+01

A missing value assessment was performed to identify data quality issues prior to preprocessing.

```
In [33]: #Missing Value Analysis
print("Temperature Missing Values")
print(temp_df.isnull().sum())

print("\nAir Quality Missing Values")
print(aq_df.isnull().sum())

print("\nGeospatial Missing Values")
print(geo_df.isnull().sum())
```

```
Temperature Missing Values
Land-Ocean: Global Means    0
dtype: int64
```

```
Air Quality Missing Values
City          0
Date          0
PM2.5        4598
PM10        11140
NO           3582
NO2          3585
NOx          4185
NH3         10328
CO           2059
SO2          3854
O3           4022
Benzene       5623
Toluene       8041
Xylene       18109
AQI           4681
AQI_Bucket    4681
dtype: int64
```

```
Geospatial Missing Values
Land          0
Year          0
Series        0
Value         0
dtype: int64
```

```

In [34]: # Temperature Trend Graph
# Convert the index (years) and J-D column to numeric

temp_plot = temp_df.copy()

temp_plot = temp_plot.reset_index()

# Rename columns for convenience
temp_plot.columns = ['Year', 'Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
                    'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec',
                    'J-D', 'D-N', 'DJF', 'MAM', 'JJA', 'SON']

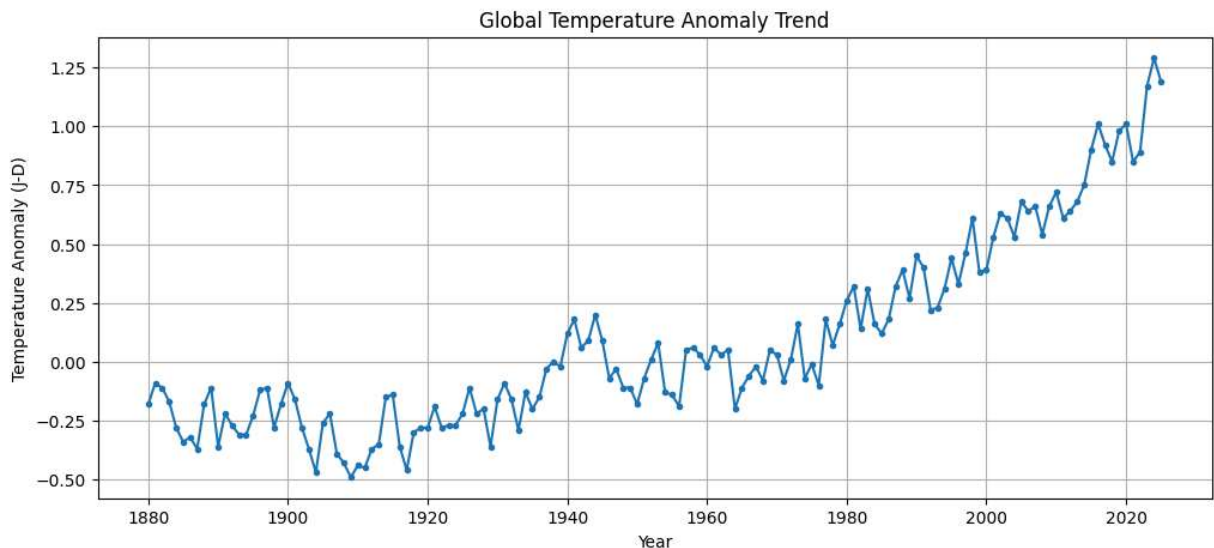
temp_plot['Year'] = pd.to_numeric(temp_plot['Year'], errors='coerce')
temp_plot['J-D'] = pd.to_numeric(temp_plot['J-D'], errors='coerce')

plt.figure(figsize=(12,5))
plt.plot(temp_plot['Year'], temp_plot['J-D'], marker='.')

plt.title('Global Temperature Anomaly Trend')
plt.xlabel('Year')
plt.ylabel('Temperature Anomaly (J-D)')
plt.grid(True)

plt.show()

```



Temperature Graph Insight: Global temperature anomalies show a clear upward trend, particularly after 1980, indicating long-term warming.

```

In [35]: #Air Quality Graph
city_aqi = (
    aq_df.groupby("City")["AQI"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

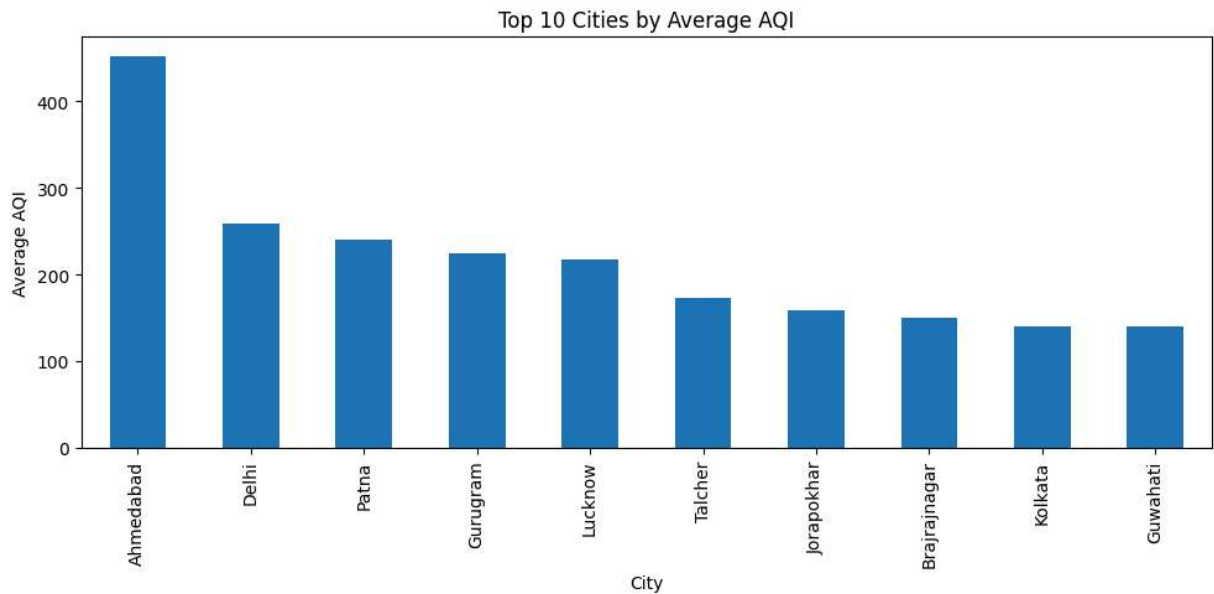
plt.figure(figsize=(10,5))

```

```
city_aqi.plot(kind="bar")

plt.title("Top 10 Cities by Average AQI")
plt.xlabel("City")
plt.ylabel("Average AQI")

plt.tight_layout()
plt.show()
```

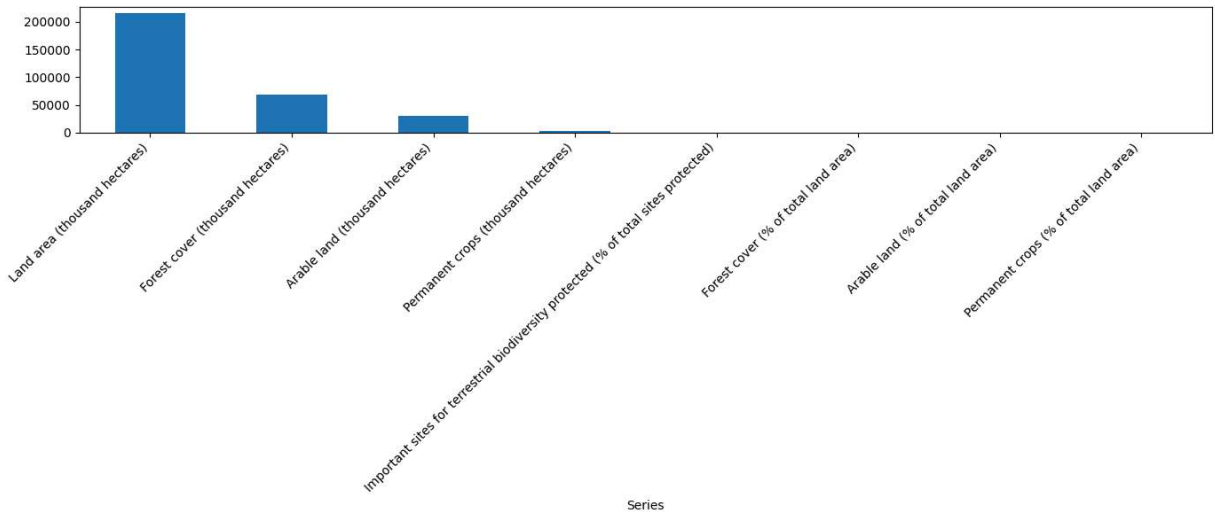


AQI Graph Insight: Ahmedabad records the highest average AQI among the top 10 cities analyzed.

```
In [36]: # Geospatial Graph

geo_summary = (
    geo_df.groupby("Series")["Value"]
    .mean()
    .sort_values(ascending=False)
    .head(10)
)

plt.figure(figsize=(14,6))
geo_summary.plot(kind='bar')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Geospatial Graph Insight: Land area and forest cover indicators contribute the largest values in the geospatial dataset.

Observations

Temperature Dataset

- The global temperature anomaly dataset shows a clear long-term warming trend from 1880 to the present.
- Temperature anomalies increased significantly after 1980.
- Recent years exhibit the highest recorded temperature anomalies, indicating accelerating global warming patterns.

Air Quality Dataset

- Significant variation in AQI values was observed across cities.
- Ahmedabad recorded the highest average AQI among the top 10 cities analyzed.
- The dataset contains several missing values that will be handled during preprocessing.

Geospatial Dataset

- Land area and forest cover indicators dominate the environmental dataset.
- Environmental indicators vary considerably across regions and land-use categories.
- The dataset appears complete with no missing values.

Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib

- Visual Studio Code

Conclusion

Week 1 focused on identifying, collecting, and performing initial exploration of environmental datasets.

Three datasets covering temperature, air quality, and geospatial indicators were successfully acquired and analyzed.

Initial exploratory analysis provided insights into dataset structure, missing values, environmental indicators, and urban air quality patterns.

These findings establish a strong foundation for data cleaning, preprocessing, and integration activities planned for Week 2.

Week 2 will focus on data cleaning, preprocessing, handling missing values, and integrating the datasets for further analysis.